

Exercise 2

Frequency response measurements

For all exercises you will need a Windows or Linux pc with Matlab/Simulink installed.

2.1 Frequency Response Function of a notch filter

A notch filter is a controller element which is meant to suppress a very specific frequency. It is defined, in continuous-time, as

$$F(s) = \frac{\frac{1}{(2\pi f_1)^2} s^2 + \frac{2\beta_1}{2\pi f_1} s + 1}{\frac{1}{(2\pi f_2)^2} s^2 + \frac{2\beta_2}{2\pi f_2} s + 1},$$

where f_1 and f_2 are the locations of the zeros and poles (in Hz) respectively, and β_1 and β_2 are the dampings of these zeros and poles. For a pure notch filter $f_1 = f_2$.

- Create a Simulink model with a notch filter, e.g. with a standard ‘Transfer Fcn’-block from the Simulink library. Apply a broadband noise (‘Random number’) as input to the filter. Place the zeros (numerator) and the poles (denominator) of the notch at 40 Hz, with a damping of 0.01 for the zeros and a damping of 1 for the poles.
- Simulate the notch over a time interval of 30 s with a fixed step solver (e.g. `ode1` or `ode3`) with a step size of 0.001 s. Save both the input and output of the filter to the Matlab workspace. Compare the filter response with its input in time domain. Do you see any difference?
- Compare the input and output in frequency domain by calculating their auto power spectral densities. What do you see now? *Hint: see Matlab functions `fft.m`, `pwelch.m` or `periodogram.m`.*
- Determine the FRF of the notch filter using the ratio between the appropriate cross and auto power spectral densities. Also, determine the corresponding coherence function. Does the outcome match your expectation? *Hint: see Matlab functions `cpsd.m` and `mscohere.m`.*
- Increase the damping of the zeros by a factor 10. How do the FRF and coherence change? Explain what you see.

2.2 Frequency Response Function of a mass system

In this exercise you will perform a closed-loop FRF measurement of a simple mass, controlled by a PD. Mathematically, this plant and controller are defined as

$$H(s) = \frac{1}{ms^2} \quad \text{and} \quad C(s) = P + Ds,$$

where s is the Laplace variable.

- a. Assume the mass is 1 kg. Using the above definitions, calculate appropriate P and D so that the bandwidth (cross-over frequency of the open loop $C(s)H(s)$) is 10 Hz. To this end, place the zero of the controller at the desired bandwidth, then adjust the controller gain so that the open-loop gain is 1 (i.e. 0 dB) at the same frequency. Demonstrate your design by making a Bode plot of the open loop $C(s)H(s)$.
- b. Create a Simulink model with a 1 kg mass and your calculated PD-controller. Inject a noise signal in the closed loop just behind the controller block.
- c. Simulate the model over 120 s with the fixed step Euler solver with step size 0.001 s. Save the relevant signals for a closed-loop sensitivity FRF measurement to the Matlab workspace. Study these signals in both time domain and frequency domain.
- d. Determine the coherence function and the FRF from the data obtained in part b. Does the outcome match your expectation?
- e. Compute the open-loop FRF from the sensitivity FRF. How does it compare to your design of $C(s)H(s)$ in part a.? Explain what happens with the phase of the FRF for high frequencies.
- f. Perform a separate FRF-measurement of the PD-controller only and determine the quotient of the open-loop FRF from part d. and the newly measured controller FRF. To what extent does the obtained result resemble a double integrator behavior?

2.3 Closed loop FRF measurement

Download the Simulink file `frf_ex3.mdl` from Canvas (this file is meant for Matlab r2006b or higher; Matlab 7.0.4 users can download `frf_ex3_m7.mdl`, Matlab 6.1 users can download `frf_ex3_m6.mdl`).

1. Open the Simulink file `frf_ex3.mdl`. This model simulates a real experiment, where the plant FRF has to be determined. It contains an unknown low-bandwidth controller, and an unknown plant. Assume that the controller cannot be measured separately. Check whether the simulation parameters are set to the right values: fixed-step solver, ode1 (Euler), simulation time of 30 seconds with a step size of 0.001.
2. Determine the sensitivity of the closed loop system by performing an FRF measurement, i.e. by inserting noise at the right place and measuring the right signals. Use the Matlab function `tfe.m` or `tfestimate.m`, and choose `nfft` such that the resolution of the frequency vector is 0.25 Hz. Choose `noverlap` to be half this size. Make sure to plot

the amplitude, the phase and the coherence of the sensitivity. Why is the coherence not equal to 1 at low frequencies? So between which frequencies is this measurement reliable?

3. Which signals have to be measured to determine the process sensitivity of the closed loop? Perform this measurement. Make sure to plot the amplitude, the phase and the coherence of the process sensitivity. What happens to the coherence at high frequencies? Can you explain this? Between which frequencies is this measurement reliable?
4. How can you determine the plant dynamics using only sensitivity and process sensitivity (without measuring the controller)? Determine the plant FRF and plot amplitude and phase. Keeping in mind that this is a simulation of a real experiment, what could cause the two high frequent phenomena: bad amplitude estimate and phase drop? How much is this phase drop and can you explain this?
5. Repeat this exercise with different values for `nfft` en different simulation times in Simulink. What differences do you notice and how can you explain these differences?